A Report on a Patriot in Our Society

Nan Rendong — The Father of China's Sky Eye

Introduction

Patriotism is not merely about waving a national flag or shouting slogans; it is about dedicating one's life to a cause that strengthens the nation and benefits its people. Among the many patriots in the field of science and technology in China, Nan Rendong (1945–2017) stands out as a remarkable example. As the chief scientist and chief engineer of the Five-hundred-meter Aperture Spherical Radio Telescope (FAST), also known as "China's Sky Eye," Nan devoted 22 years of his life to building the world's largest and most sensitive single-aperture radio telescope. His story is one of unwavering patriotism, relentless perseverance, and selfless dedication to the nation's scientific advancement.

Part I: What Nan Rendong Has Done and Achieved

1.1 Giving Up Comfort Abroad to Serve the Motherland

In the early 1990s, Nan Rendong was working as a visiting professor at the National Astronomical Observatory of Japan, enjoying a prestigious position and a generous salary — his daily wage in Japan exceeded what he earned in an entire year in China. However, when the idea of building a new generation of large radio telescopes was proposed at the 1993 International Radio Science Union meeting in Tokyo, Nan saw a once-in-a-lifetime opportunity for China. He made a decisive choice: he would return home and lead the effort to build China's own world-class telescope, refusing to let the country fall behind in the global astronomical community.

1.2 A 12-Year Search for the Perfect Site

Building a 500-meter telescope required finding a natural depression of the right shape and size — a karst valley that could house the massive structure. Starting in 1994, Nan led his team on a grueling site-selection journey across the mountains of Guizhou Province in southwestern China. Carrying over 300 satellite remote-sensing images, he trudged through rugged terrain for more than a decade, examining hundreds of candidate valleys. He once narrowly escaped being swept away by a flash flood during a field survey. After evaluating over 3,000 depressions, the team finally identified "Dawodang" — a perfect karst sinkhole in Pingtang County — as the ideal location in 2006. The entire site selection process took 12 years of painstaking effort.

1.3 Overcoming Unprecedented Technical Challenges

With no existing blueprint to follow — the international community imposed a complete technology blockade on China — Nan and his team had to innovate everything from scratch. One of the most daunting challenges came in 2010, just before construction began, when all prior cable-net experiments had failed. The steel cables used in standard suspension bridges snapped under FAST's fatigue tests. Nan refused to give up, declaring: "We have no way out — we must try again!" After nearly a hundred failures, the team successfully developed cables with a tensile strength of 500 MPa and a fatigue resistance of 2 million stretches — 2.5

times the national standard. Every component of FAST, from the cable-net structure to the feed cabin system, was independently developed by Chinese engineers.

1.4 Completing the World's Greatest Telescope

On September 25, 2016, after 22 years of planning and construction, FAST was officially completed and began receiving electromagnetic signals from the depths of the universe. With a reflector as large as 30 football pitches and 4,450 panels, FAST is capable of detecting signals from 13.7 billion light-years away. Its sensitivity is approximately 10 times that of the American Arecibo telescope, making it the world's leading radio telescope for the next 20 to 30 years. Just weeks after Nan's passing, FAST captured its first pulsar signal from deep space — a poetic tribute to the man who gave his life to see the universe.

1.5 Fighting Illness Until the Very End

In 2015, at the age of 70, Nan was diagnosed with late-stage lung cancer. Surgery damaged his vocal cords, leaving him with a hoarse voice. Yet just three months after the operation, he returned to the construction site, pushing himself to monitor the project's progress. Even from his hospital bed in his final days, he continued to call colleagues about technical issues. He passed away on September 15, 2017. In 2018, the International Astronomical Union named Asteroid 79694 "Nanrendong" in his honor. In 2019, he was posthumously awarded the national honorary title of "People's Scientist." In 2025, his story was included in China's eighth-grade Chinese language textbook under the title "There Is a Star Named Nan Rendong."

Part II: Inspirations from Nan Rendong's Story

2.1 Acquiring Cutting-Edge Knowledge to Stay Internationally Competitive

Nan Rendong's greatest legacy is his insistence on independent innovation. When the international community imposed technology blockades and skeptics dismissed the project as "outdated," he firmly believed that a telescope more powerful than the Nobel Prize-winning Arecibo could never be outdated. He and his team solved every technical problem from scratch — from the cable-net design to the feed cabin system — proving that self-reliance is the only path to genuine scientific strength. This inspires me to acquire cutting-edge knowledge and skills, not merely to pass exams, but to contribute to China's capacity for independent innovation. In an era of increasing global competition and technological containment, the ability to innovate independently is not optional — it is a matter of national survival.

2.2 Choosing a Career That Benefits Society

Nan's decision to abandon a comfortable life abroad and return to China resonates deeply with me as a future educator. He did not choose the easiest path; he chose the path where he was most needed. His dedication to building a tool that would serve not just himself but the entire nation — and indeed the global scientific community — reminds me that a meaningful career is one where personal ambition aligns with social need. As a student preparing to become a teacher, I am inspired to seek not just personal success, but a career where my knowledge can benefit society. Nan's story reinforces my belief that education is itself a form of nation-building — equipping the next generation with the ability to think critically and innovate independently.

2.3 Perseverance in the Face of Adversity

Perhaps the most personal lesson from Nan's story is his extraordinary perseverance. He spent 12 years just finding the right location. He endured nearly a hundred failed experiments before solving the cable problem. He continued working even as cancer consumed his body. His motto — "If the Sky Eye has even a single flaw, we would be unworthy of our country" — speaks to a standard of excellence driven not by ego but by a sense of duty. This teaches me that true achievement is never quick or easy; it requires the willingness to endure hardship, to persist when others have given up, and to hold oneself to the highest standard not for personal glory but for the people one serves.

2.4 The Spirit of Selfless Patriotism

Nan Rendong's patriotism was not performative. He did not give speeches about loving his country; he simply devoted 22 years of his life to building something that would make China stronger. He wore work clothes, lived simply, and described himself as "a tactical old worker, not a strategic master." His patriotism was expressed through action — through sweat, through sleepless nights, through a willingness to sacrifice personal comfort for a greater cause. This challenges me to reflect on what patriotism truly means in my own life: not grand declarations, but the daily choice to contribute what I can, where I am, with what I have.

Conclusion

Nan Rendong's life is a testament to the power of one person's dedication to transform a nation's scientific landscape. He gave up wealth and comfort abroad, endured decades of hardship in the mountains of Guizhou, overcame impossible technical challenges, and fought illness to the very end — all so that China could have its own eye on the universe. His story teaches us that true patriotism is measured not in words but in a lifetime of action, that knowledge acquires its highest value when it serves the nation and humanity, and that perseverance in the face of adversity is not just a personal virtue but a national asset. As we look up at the night sky and the star now bearing his name, we are reminded that the greatest patriots are those who build something that outlasts them — something that continues to illuminate the world long after they are gone.